

Design-to-Delivery Accelerator Challenge

A Prompt Intelligence Engine for brand-consistent,
compliant web delivery at scale **powered by data science**.

Link: <https://prompt-muse-ai-75.vercel.app>

Access Key: judge_access

The Real Problem Pfizer Faces

Criterion 1 — Insight & Problem Understanding

2,000+

websites to manage

80

countries, each with
regulatory variations

weeks

average time to launch
a single new page

manual

compliance review,
slide by slide

The Gap

- ✗ Prompts sent to LLMs with no quality signal
- ✗ Generic output — not audience or market specific
- ✗ Compliance checked after generation, not before
- ✗ No audit trail — humans can't explain AI decisions
- ✗ Brand drift across 80 markets with no measurement

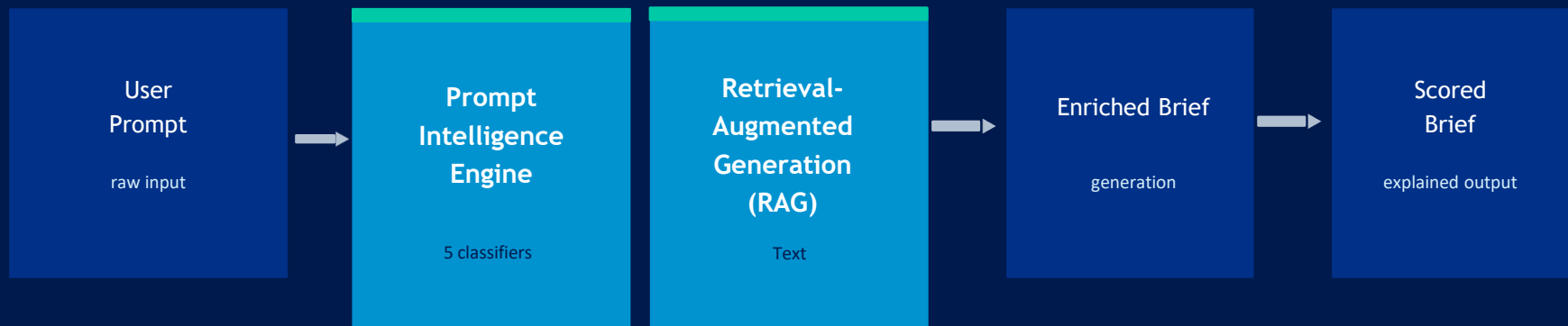
These gaps compound at 2,000+ sites.

The cost is inconsistency, risk, and wasted human review time.

Approach

Treat the prompt as a signal, not a string.

Before any LLM sees a brief, a data-science pipeline classifies, scores and enriches it — so what goes in is already intelligent.



This is where Data Science adds value — measurable, explainable, repeatable.

Access Key = judge_access

[←](#) [→](#) [↻](#) <https://prompt-muse-ai-75.vercel.app/> [Sharing with Agent](#) [🗨](#) [📄](#) [📧](#) [⋮](#)

Company Name | Design-to-Delivery Accelerator · Prompt Intelligence Engine CXI+AI · University of Liverpool

Welcome to the Accelerator

Enter your details to personalise your workspace.

All fields are mandatory.

FIRST NAME	LAST NAME
Jane	Smith

COMPANY EMAIL

PHONE NUMBER

EMPLOYEE NUMBER	DEPARTMENT
PFZ-00000	Digital Marketing

COUNTRY / REGION

The Prompt Intelligence Engine

Criterion 2 — Creative Use of AI in the Web Design Lifecycle

01

Audience Classifier

PIE runs as a protected Supabase Edge Function (pie-classify) Outputs: audience, jurisdiction, risk, tone, readability, PIE score

Supabase Edge Function

02

Jurisdiction Detector

Maps country to regulatory framework (EMA, FDA, MHRA, TGA)

Request auth enforced with x-judge-access-key

auth: x-judge-access-key

03

Content Risk Scorer

Flags sensitive terms (clinical claims, pricing)
Outputs 0–1 risk score

risk + tone classifier

04

Brand Tone Analyser

Jurisdiction-aware and risk-aware prompt enrichment
Enriched prompt passed to generation pipeline

lexical RAG retrieval

05

Readability Predictor

RAG context injected from curated compliance corpus
Lexical retrieval path in live system

Groq LLM

Sample Engine Output

Input: "page for our oncology drug for doctors in Germany"

Audience

HCP

Jurisdiction

DE → EMA

Risk Score

0.87 HIGH

Brand Tone

Clinical

FK Grade

11 (HCP: ✓)

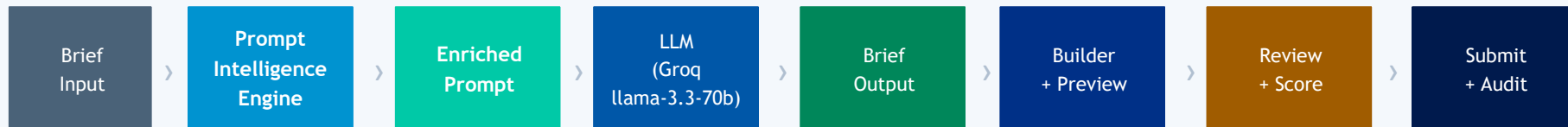
PIE Score

83 / 100

Every decision is logged. No opaque scoring.

Technical Architecture

Criterion 3 — Technical Execution & Integration



Live Production Stack

- Frontend: React + Vite + TypeScript (Vercel)
- Backend: Supabase Edge Functions
- Functions: pie-classify, generate-brief, review-content, notify-reviewer, analyze-brief
- LLM provider: Groq (llama-3.3-70b-versatile)
- RAG: policy corpus retrieval with lexical fallback
- Access control: x-judge-access-key enforced on all endpoints

Why This Architecture

- Classifier-first pipeline — enrichment before generation
- Unauthorized or invalid key blocked before AI processing
- Each classifier output is traceable — explainable, not opaque
- End-to-end tested: portal, PIE, brief generation, submission flow
- Audit log generated on every brief run — traceable outputs per stage
- Cloud-first, secured, and auditable — deployed for judging

Trust, Compliance & Explainability

Criterion 4 — Trust, Compliance & Explainability

Every score has a reason

Protected endpoints require judge access key.
Unauthorized or invalid key is blocked before AI processing. Risk score 0.87 because: keyword 'oncology drug' detected + jurisdiction Germany — EMA Directive 2001/83/EC applies. Not just a number — a traceable chain.

Audit trail built in

Every brief run logs: timestamp, classifier outputs, prompt version, LLM used, review decisions. Traceable outputs returned per stage (PIE, brief generation, review). 2,000 sites x accountability = governance that scales.

Human review gates

High-risk prompts (score > 0.75) are flagged before the LLM runs. Humans confirm or override. The AI suggests — humans decide.

90% threshold, explained

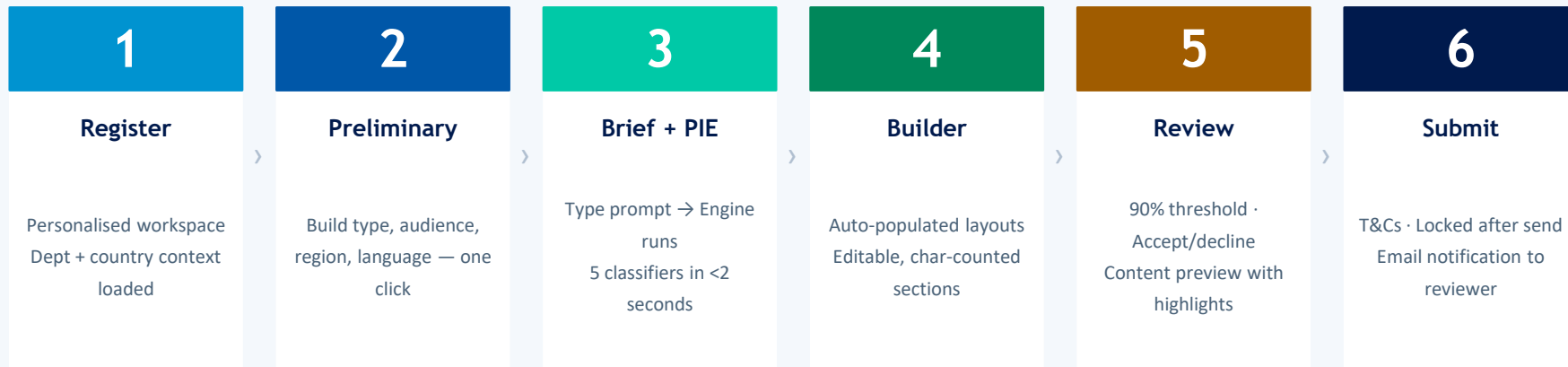
Submissions below 90% score are blocked, with specific recommendations: not 'failed' but 'here is the exact change needed and why it matters.'

Regulatory Coverage

Jurisdiction	Framework	Auto-detected
EU (Germany)	EMA Dir. 2001/83/EC	✓
USA	FDA 21 CFR	✓
UK	MHRA/ABPI	✓
Australia	TGA	✓
Global	WCAG 2.1 AA	✓
All markets	GDPR	✓

User Experience & Workflow Clarity

Criterion 5 — UX & Workflow Clarity



Designed for multiple roles — one interface, clear workflow for each

Content Author	Designer	QA / Compliance	IT Reviewer
Natural language prompt → brief built automatically. No technical skill required.	6 Canva-style layout templates. Live editable canvas with char counts.	Automated checks with pass/fail + specific fix recommendations. 90% gate.	Submission locked after send. Audit trail. Email notification. No chasing.

The Data Science Advantage

What is the difference?

	Typical Submission		This Submission
Brief handling	<i>String → LLM</i>	●	Signal → Classifier → LLM
Compliance check	<i>After generation</i>	●	Before generation
Score explanation	<i>Opaque number</i>	●	Classifier confidence + reason
Audience detection	<i>Manual / assumed</i>	●	Classifier-first pipeline (pie-classify endpoint)
Readability	<i>Guessed</i>	●	Predicted (Groq-backed, FK grade enriched)
Jurisdiction	<i>Not handled</i>	●	Auto-mapped to regulatory body
Audit trail	<i>None</i>	●	Full log, every decision
Data science used	<i>Prompt engineering</i>	●	ML pipeline + lexical RAG + Groq generation

"Most tools treat the prompt as a string.

We treat it as a signal.

Before any LLM sees a brief, our pipeline classifies audience, jurisdiction, risk, tone and readability -- with confidence scores and audit trails.

The LLM gets a better prompt. The output is consistent.

Compliance teams have evidence, not hope.

That's data science applied to the governance problem at scale."